



Methods

Preview

1. Data: acoustic/articulatory data from published datasets
2. Distinctiveness measurement in acoustics & articulation
- 3. Bigram surprisal calculation**
4. Phoneme flexibility: degree of freedom between articulatory synergies & the speech target



USC



Methods

Bigram contextual surprisal

- Probability estimated as frequency in corpora:
 - British English: the British National Corpus (BNC Consortium, 2007)
 - American English: the Open American National Corpus
 - French: French-Public Domain-Book Corpus (Langlais et al. 2024)
- Included both segmental & syllabic information

$$I(\text{bigram} \mid \text{context}) = -\log(P(\text{bigram} \mid \text{context}))$$

